

The project was designed as a complete omnichannel ecosystem consisting of:

- A customer-facing Web Application
- Native-like Mobile Applications for Android & iOS
- AI-powered Smart Kiosk Applications deployed inside retail stores
- Cloud-based Admin & Retail Management System
- Centralized Analytics & Recommendation Engine

The core objective of OUI was to eliminate the traditional friction of physically trying clothes in stores while improving customer convenience, reducing waiting time, and increasing operational efficiency for fashion retailers.

Using AI-powered virtual try-on technology, customers can stand in front of an interactive smart kiosk inside a clothing store and instantly visualize how garments look on their body without entering trial rooms or physically wearing multiple outfits.

The ecosystem bridges physical retail and digital commerce by creating a seamless shopping journey where customers can:

- Browse apparel collections digitally
- Virtually try garments in real time
- Save outfits to profiles or wishlists
- Purchase products instantly
- Continue shopping later from mobile or web applications
- Receive personalized recommendations
- Share looks socially

The platform was architected with scalability, real-time rendering, cloud synchronization, and multi-device interoperability at its core, enabling fashion brands to deploy the solution across multiple stores and regions with centralized management.

2. Industry Problem Statement

The fashion retail industry has traditionally relied on physical trial rooms as the primary method for customers to evaluate clothing before purchasing. While this approach has existed for decades, it introduces multiple operational and customer experience challenges.

Existing Industry Challenges

Long Waiting Times in Trial Rooms

Customers often spend significant time waiting for fitting rooms during peak hours, leading to frustration and abandoned purchases.

OUI – Cloth Virtual Try-On Ecosystem

Project Case Study

Platform: Web Application | iOS Application | Android Application | Smart Kiosk Application

Industry: Fashion Retail Technology / Retail Experience Transformation

Architecture: Scalable Microservice-Oriented Architecture

Technology Partner: Corescent Technologies

Deployment Model: Cloud-Native & Multi-Platform Ecosystem

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1. Project Overview

OUI – Cloth Virtual Try-On is a next-generation fashion retail ecosystem developed by Corescent Technologies to transform how customers interact with clothing brands inside retail stores and across digital channels.

- Reduce customer waiting time in retail stores
- Improve customer engagement and shopping experience
- Increase purchase conversion rates
- Digitize the in-store apparel discovery process
- Reduce dependency on physical trial rooms
- Enable data-driven merchandising decisions
- Build a connected retail ecosystem across kiosk, mobile, and web
- Increase operational efficiency for fashion brands

Technical Objectives

- Build a highly scalable multi-platform architecture
- Enable real-time virtual try-on rendering
- Support cloud synchronization across devices
- Ensure low-latency image processing and garment rendering
- Build secure and scalable APIs
- Provide centralized product and inventory management
- Support future AI model enhancements
- Enable analytics-driven recommendations

Customer Experience Goals

- Deliver a frictionless shopping journey
- Allow instant virtual clothing visualization
- Provide personalized outfit recommendations
- Enable cross-device continuity
- Reduce shopping fatigue
- Make fashion exploration more interactive and enjoyable

4. Solution Strategy & Product Ecosystem

Corescent Technologies designed OUI as a fully integrated retail technology ecosystem rather than just a standalone virtual try-on application.

The ecosystem consists of multiple interconnected modules:

Platform	Purpose
Smart Kiosk Application	In-store AI-powered virtual try-on experience
Mobile Applications	Personalized shopping, saved looks, purchases

Web Application	Online browsing, management, analytics
Admin Dashboard	Centralized operational control
Retail Management Portal	Inventory, stores, campaigns, analytics
Cloud Backend Infrastructure	APIs, authentication, rendering orchestration
AI Processing Engine	Body detection, garment mapping, recommendations

Omnichannel Shopping Experience

The ecosystem allows customers to start their shopping journey on one platform and continue seamlessly on another.

Example:

1. Customer visits store
2. Uses kiosk for virtual try-on
3. Saves favorite outfits
4. Receives recommendations in mobile app
5. Completes purchase online later
6. Receives personalized future suggestions

This omnichannel continuity dramatically improves customer retention and engagement.

5. Core Features & Functional Modules

AI-Powered Virtual Try-On

The centerpiece of the ecosystem is the intelligent virtual try-on engine.

Customers standing in front of the kiosk can:

- Detect body posture automatically
- Visualize garments in real time
- Change apparel instantly
- Compare multiple outfits
- View different colors and sizes
- Capture and save looks
- Share styles digitally

Smart Kiosk Experience

The kiosk acts as an interactive digital mirror inside retail stores.

Features include:

- Gesture-enabled interactions
- Touchscreen support
- Real-time body tracking
- Product scanning
- Outfit recommendations
- QR-based session transfer to mobile
- Digital cart creation
- Instant store inventory lookup

Mobile Applications (Android & iOS)

The mobile applications extend the in-store experience.

Features include:

- Login & profile management
- Saved virtual outfits
- Personalized recommendations
- Product browsing
- Wishlist & favorites
- Shopping cart
- Order management
- Loyalty integration
- Push notifications
- Cross-device synchronization

Web Application

The web platform provides:

- Full product catalogue browsing
- Virtual try-on support
- Retail analytics dashboards
- User management
- Store management
- Campaign management
- Recommendation configuration
- Product onboarding

AI Recommendation Engine

The platform intelligently suggests:

- Matching outfits
- Trending apparel
- Similar products
- Personalized style recommendations
- Seasonal suggestions
- Cross-selling opportunities

Retail Analytics Dashboard

Brands and store operators can monitor:

- Customer engagement metrics
- Most tried products
- Popular categories
- Conversion analytics
- Heatmaps of interactions
- Session analytics
- Store performance
- Inventory trends

Real-Time Inventory Integration

The platform integrates with inventory systems to:

- Show live stock availability
 - Prevent out-of-stock purchases
 - Sync store inventory
 - Enable omnichannel fulfillment
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6. Kiosk Experience Workflow

The kiosk application was one of the most innovative components of the project.

Step 1: User Detection

As customers approach the kiosk, the system activates body and motion detection.

Step 2: Body Mapping & Calibration

Computer vision models identify:

Time-Consuming Shopping Experience

Trying multiple garments physically consumes time and energy, reducing customer satisfaction and limiting the number of products explored.

Inventory Handling Overhead

Store staff constantly manage clothing movement between racks, trial rooms, and inventory returns, increasing operational inefficiency.

Hygiene & Convenience Concerns

Many customers avoid physically trying clothes due to hygiene concerns, especially in high-traffic retail environments.

Limited Customer Engagement

Traditional stores provide little personalization or digital interaction, making it difficult to create immersive shopping experiences.

Lack of Data Insights

Retailers often lack visibility into:

- Which products customers virtually explored
- What styles customers preferred
- Which garments had high engagement but low conversion
- Customer behavioral analytics inside physical stores

High Operational Costs

Managing trial rooms, staffing, garment organization, and customer support increases operational expenditure for retail businesses.

3. Vision, Goals & Objectives

Vision

To redefine the future of fashion retail by combining AI, computer vision, cloud technology, and immersive digital experiences into a unified ecosystem that makes shopping faster, smarter, and more engaging.

Business Objectives

- Body posture
- Body dimensions
- Skeleton mapping
- Pose orientation

Step 3: Apparel Selection

Customers browse clothing collections directly from the kiosk interface.

Step 4: Virtual Garment Rendering

Selected garments are digitally overlaid on the customer in real time.

Step 5: Interaction & Recommendations

The system recommends:

- Matching products
- Similar styles
- Accessories
- Alternate colors

Step 6: Save & Continue

Users can:

- Save looks
- Generate QR codes
- Transfer sessions to mobile apps
- Add products to cart
- Share outfits socially

This workflow significantly reduces dependency on physical fitting rooms while improving shopping speed and convenience.

7. Web & Mobile Application Experience

Unified Customer Profiles

All platforms are connected through centralized customer profiles.

This enables:

- Saved try-on history
- Purchase continuity
- Wishlist synchronization
- Recommendation personalization
- Loyalty integration

Responsive UI/UX Design

The interfaces were designed with:

- Minimalist modern UI
- Fast navigation
- Accessibility-first approach
- Cross-device consistency
- Smooth animations
- Real-time updates

Social & Sharing Features

Customers can:

- Share outfits
- Send style recommendations
- Save fashion collections
- Collaborate on looks

Personalized Shopping Journey

The ecosystem uses behavioral analytics to personalize:

- Homepage recommendations
 - Trending suggestions
 - Seasonal collections
 - Store campaigns
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8. AI & Virtual Try-On Technology

The virtual try-on engine developed by Corescent Technologies combines multiple advanced technologies.

Computer Vision

Used for:

- Human detection
- Body segmentation
- Pose estimation
- Movement tracking
- Garment alignment

AI Garment Mapping

The engine dynamically maps garments to body posture and dimensions.

Real-Time Rendering

Optimized rendering pipelines ensure:

- Low latency
- Smooth transitions
- Realistic overlays
- High frame-rate rendering

Recommendation Intelligence

Machine learning algorithms analyze:

- Customer behavior
- Product interactions
- Purchase history
- Style preferences

Cloud-Based AI Processing

Heavy AI computations were designed to scale horizontally through cloud infrastructure.

This enables:

- Multi-store deployments
 - Simultaneous sessions
 - Real-time synchronization
 - Elastic resource scaling
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9. Architecture & Scalability

Scalable Cloud-Native Architecture

Corescent Technologies designed the system with a scalable distributed architecture capable of supporting:

- Thousands of concurrent users
- Multi-store kiosk deployments
- Real-time synchronization
- High-volume media processing
- AI workload scalability

Architectural Components

Layer	Responsibility
Presentation Layer	Web, mobile, and kiosk interfaces
API Gateway Layer	Centralized request routing
Authentication Layer	Secure identity management
AI Processing Layer	Body detection & rendering
Business Logic Layer	Product, orders, analytics
Data Layer	Databases & storage systems
Analytics Layer	Customer insights & reporting
Cloud Infrastructure Layer	Scalability & orchestration

Microservice-Oriented Backend

The backend services were modularized into independent services including:

- Authentication Service
- Product Service
- Recommendation Service
- Inventory Service
- Analytics Service
- Notification Service
- Media Processing Service
- AI Rendering Service

This architecture allowed:

- Independent deployment
- Horizontal scaling
- Faster updates
- Better fault isolation
- Easier maintenance

API-First Infrastructure

All applications communicate through centralized secure APIs enabling:

- Device interoperability
- Real-time sync
- Multi-platform support
- Future extensibility

Cloud Infrastructure

The platform was deployed using scalable cloud infrastructure with:

- Auto-scaling compute resources
- CDN media delivery
- Distributed storage
- Load balancing
- Monitoring & observability
- Disaster recovery mechanisms

Real-Time Synchronization

The ecosystem supports real-time synchronization across:

- Kiosks
- Mobile apps
- Web platforms
- Admin dashboards

Ensuring consistent user experiences across all touchpoints.

10. Development Process & Execution Strategy

Cloud & DevOps

Technology	Purpose
Docker	Containerization
Kubernetes	Scalable orchestration
CI/CD Pipelines	Automated deployments
Cloud Monitoring Tools	Infrastructure observability
Load Balancers	Traffic distribution

13. Security, Performance & Optimization

Security Measures

The platform implemented enterprise-grade security including:

- JWT-based authentication
- Role-based access control
- Encrypted API communication
- Secure media access
- Data protection standards
- Cloud security best practices

Performance Optimization

The system was optimized using:

- CDN-based media delivery
- Lazy loading strategies
- AI model optimization
- Caching layers
- GPU-assisted rendering
- Database indexing

Reliability & Monitoring

Monitoring systems track:

- Kiosk health

Discovery & Research

The project began with:

- Retail workflow analysis
- Customer behavior research
- Store operations analysis
- Technology feasibility studies

UI/UX Design Phase

Design teams created:

- Interactive prototypes
- Kiosk workflows
- Mobile journeys
- Responsive web layouts
- Accessibility standards

Backend Engineering

The backend systems were designed to support:

- High concurrency
- AI processing workloads
- Real-time synchronization
- Multi-tenant store management

AI Model Integration

The AI layer required:

- Pose estimation optimization
- Rendering accuracy improvements
- Real-time processing optimization
- Edge-device calibration

Multi-Platform Deployment

The solution was deployed across:

- Retail kiosks
- Android devices
- iOS devices
- Web browsers

- Admin portals
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11. Challenges Faced & Solutions Implemented

Real-Time Rendering Performance

Challenge

Rendering garments in real time while maintaining smooth interaction and low latency was technically demanding.

Solution

Corescent Technologies optimized:

- Rendering pipelines
 - AI model inference
 - GPU acceleration
 - Media compression
 - Edge processing workflows
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Cross-Device Synchronization

Challenge

Ensuring consistent sessions across kiosk, web, and mobile devices.

Solution

Implemented centralized session orchestration with cloud synchronization and tokenized device linking.

Store Environment Variations

Challenge

Different lighting conditions and kiosk placements affected body detection accuracy.

Solution

Adaptive calibration algorithms and dynamic computer vision tuning improved environmental reliability.

High Scalability Requirements

Challenge

The platform needed to support multiple stores and large-scale concurrent sessions.

Solution

Implemented microservice-based cloud infrastructure with auto-scaling and distributed workloads.

AI Processing Costs

Challenge

AI rendering and processing workloads can become computationally expensive.

Solution

Hybrid processing strategies were introduced combining edge optimization and cloud scalability.

12. Technologies Used

Frontend Technologies

Technology	Purpose
React.js / Next.js	Web application development
Flutter	Cross-platform mobile applications
Dart	Mobile application language

TypeScript	Scalable frontend engineering
Tailwind CSS	Modern responsive UI styling
WebSocket Integration	Real-time synchronization

AI & Computer Vision

Technology	Purpose
TensorFlow	AI model execution
OpenCV	Computer vision processing
Pose Estimation Models	Body tracking
AI Recommendation Models	Personalized suggestions

Backend Technologies

Technology	Purpose
Node.js	Backend services
Express.js / NestJS	API layer
REST APIs	Platform communication
GraphQL	Optimized data fetching
Redis	Real-time caching
RabbitMQ / Kafka	Event-driven processing

Database & Storage

Technology	Purpose
PostgreSQL	Relational data management
MongoDB	Flexible product & analytics storage
Cloud Object Storage	Media management
CDN Infrastructure	Fast global media delivery

Developed by Corescent Technologies, the platform transformed traditional retail experiences into intelligent, interactive, and highly scalable digital ecosystems.

By combining kiosk-based virtual try-on experiences with synchronized mobile and web applications, the project created a future-ready omnichannel retail platform that benefits both businesses and consumers.

The solution not only saves customer time and improves convenience but also empowers retailers with operational efficiency, advanced analytics, and scalable digital transformation capabilities.

OUI stands as a strong example of how technology-driven innovation can redefine customer engagement and shape the future of smart retail.

Built by Corescent Technologies

AI-Powered Retail Innovation | Virtual Try-On Ecosystem | Scalable Smart Retail Architecture

- API performance
 - AI processing latency
 - Error rates
 - User analytics
 - Infrastructure metrics
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14. Business Impact & Results

Retail Efficiency Improvements

The platform significantly improved operational efficiency for retailers.

Key Benefits for Businesses

- Reduced dependency on trial rooms
- Faster customer servicing
- Better inventory visibility
- Increased customer engagement
- Improved product discovery
- Higher conversion opportunities
- Reduced operational overhead

Customer Experience Transformation

Customers experienced:

- Faster shopping journeys
- Interactive digital experiences
- Personalized recommendations
- Reduced physical effort
- Seamless omnichannel engagement

Data-Driven Retail Intelligence

Retailers gained valuable insights into:

- Customer behavior
- Product engagement
- Conversion trends
- Store performance
- Fashion preferences

Scalability for Enterprise Retail

The architecture enables brands to:

- Deploy across multiple stores
 - Manage centralized operations
 - Scale infrastructure dynamically
 - Expand globally
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15. Why This Solution Matters

OUI is more than a virtual try-on system — it represents the future evolution of retail shopping.

Importance for Businesses

Operational Efficiency

The solution reduces time spent managing trial rooms, garment handling, and customer waiting queues.

Enhanced Customer Engagement

Interactive experiences increase customer retention and brand loyalty.

Omnichannel Commerce

The platform seamlessly connects physical retail with digital commerce.

Better Decision Making

Retail analytics help brands optimize inventory, merchandising, and campaigns.

Competitive Differentiation

Retailers adopting smart retail technology gain a significant market advantage.

Importance for Customers

Time Saving

Customers can try multiple outfits instantly without physically changing clothes.

Convenience

Shopping becomes faster, easier, and more enjoyable.

Personalized Experience

AI-driven recommendations create tailored shopping journeys.

Digital Continuity

Users can continue their shopping journey across devices.

16. Future Scope & Enhancements

Advanced 3D Virtual Try-On

Future versions may support fully immersive 3D body avatars.

AR Glass Integration

Support for augmented reality smart mirrors and wearable devices.

AI Stylist Assistant

AI-generated outfit recommendations and fashion consultations.

Voice-Controlled Kiosk Navigation

Hands-free interactions for enhanced accessibility.

Hyper-Personalization

Advanced AI personalization based on style patterns and preferences.

Social Commerce Features

Influencer integrations and collaborative fashion sharing.

Smart Store Analytics

Deeper AI-driven customer behavior analysis.

Global Retail Expansion

Scalable architecture allows rapid onboarding of global fashion brands.

17. Key Learnings

Omnichannel Experiences Are the Future

Customers expect seamless transitions between physical and digital retail.

AI Must Be Practical

Technology adoption succeeds when it solves real customer pain points.

Scalability Must Be Planned Early

Building scalable architecture from the beginning reduces long-term complexity.

Real-Time Performance Is Critical

User engagement depends heavily on responsiveness and rendering speed.

Retail Data Is Extremely Valuable

Analytics provide retailers with strategic business intelligence.

18. Conclusion

OUI – Cloth Virtual Try-On successfully demonstrates how AI, cloud computing, computer vision, and scalable digital architecture can revolutionize the fashion retail industry.